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RISK AND MONEY MANAGEMENT

Maximum Adverse Excursion (MAE) and Maximum Favorable Excursion (MFE) Explained



By Oddmund Groette | April 5, 2025 | Risk and money management

How many times have we witnessed in our trading systems how trades move against our position? Many! Well, in this article, we will see how to analyze this behavior in our trades and what conclusions we can draw from them to improve our strategies. **What**

are Maximum Adverse Excursion and Maximum Favorable Excursion?

Maximum Adverse Excursion (MAE) is a term used in trading and refers to the maximum amount of loss experienced by a trade from its entry point to its maximum loss point before the trade finally turns in favor of your position or is closed. Opposite, **Maximum Favorable Excursion (MFE)** refers to the maximum amount of profit that is reached during a trade before the price retraces and the trade is finally closed.



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John Sweeney, Technical Editor of *Technical Analysis of Stocks and Commodities* magazine, introduced the concept of Maximum Adverse Excursion, which was designed to help traders determine the appropriate stop level based on historical testing. Essentially, this strategy evaluates each trade to determine the [level of drawdown](#) at which trades typically do not recover. All systems have some drawdown, and MAE aims to differentiate between normal and abnormal drawdown levels.

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Key takeaways

- The article introduces two key trading metrics: Maximum Adverse Excursion (MAE) and Maximum Favorable Excursion (MFE).
- **Maximum Adverse Excursion (MAE):** This metric quantifies the maximum loss a trade experiences from its entry point before it either turns profitable or is closed. For instance, in a long position, MAE is calculated as the difference between the entry price and the lowest price reached before the trade becomes favorable.
- **Maximum Favorable Excursion (MFE):** Conversely, MFE measures the maximum profit attained during a trade before the price retraces and the trade is closed. It represents the peak unrealized profit during the lifespan of a trade.
- These concepts were introduced by John Sweeney, the Technical Editor of *Technical Analysis of Stocks and Commodities* magazine, to assist traders in determining appropriate stop-loss levels based on historical data. By analyzing MAE and MFE, traders can identify typical drawdown levels, helping to differentiate between normal and abnormal price movements.
- Understanding these metrics allows traders to set stops and targets that align with the historical behavior of their trading instruments, potentially improving the overall performance of their strategies.

How do you calculate Maximum Adverse Excursion (MAE)?



MAXIMUM ADVERSE EXCURSION (MAE) AND MAXIMUM FAVORABLE EXCURSION (MFE)

What are MAE and MFE?

- **MAE (Maximum Adverse Excursion):** Maximum loss a trade incurs before turning profitable (Long) or minimum gain before turning unprofitable (Short).
- **MFE (Maximum Favorable Excursion):** Maximum profit a trade reaches before closing.

How to calculate MAE (example)

- Long trade: $MAE = \text{Entry price} - \text{Lowest price during the trade (before it goes up)}$.
- Short trade: $MAE = \text{Entry price} - \text{Highest price during the trade (before it goes down)}$.

How are they used?

- **Evaluate risk-reward:** If MAE is much higher than MFE, the risk outweighs potential reward (adjust exit strategy).
- **Set stop-loss levels:** Use MAE to set realistic stop-loss levels to limit risk.
- **Identify profit opportunities:** Use MFE to see if profits were maximized on past trades (adjust exit strategy).
- **Improve risk management:** MAE and MFE help manage risk by setting stop-loss levels and capitalizing on profit opportunities.

Important Note

- Larger sample size is needed for accurate MAE calculation (to capture a wider range of market conditions).

Maximum-Adverse-Excursion MAE

MAE (Maximum Adverse Excursion) is particularly useful for analyzing a set of trades. This way, an asset's price behavior is studied after entry.

Let's take a long position as an example: The entry point is the [Open Price](#), and the difference between this Open Price and the minimum or maximum Low recorded during the operation until it changed in our favor is our Maximum Adverse Excursion.

Conversely, during a Short operation, the Maximum Adverse Excursion will be the difference between our Open Price and the maximum or high reached by the price before turning in the opposite direction.

What is Maximum Favorable Excursion (MFE)?

As you have probably already deduced, Maximum Favorable Excursion (MFE) is another term used in trading. It refers to the maximum amount of profit that is reached during trade before the price retraces and the trade is finally closed. In other words, it is the

maximum amount of money a trader could have earned in a trade if they had exited at the optimal moment.

Examples of Maximum Adverse Excursion (MAE) and Maximum Favorable Excursion (MFA)

Here is a graphical example of MAE:



MAE

Here is a graphical example of MFE:



MFE

What are Maximum Adverse Excursion (MAE) and Maximun Favorable Excursion (MFE) in trading?

Experienced traders learn that as important as it is to have an effective method to determine when to trade, it is equally essential to develop a methodology to determine how much to risk.

A trader that risks too much, increases the chance that they will not survive long enough to realize the long-run benefits of a valid trading strategy.

However, risking too little creates the possibility that a trading methodology may not realize its' full potential. MAE and MFE are risk management tools for traders.

These metrics can be expressed in various ways:

- Pips
- \$ dollar per trade
- Multiples of Risk

- % percentage

MAE and MFE improve your stop losses and profit targets

Maximum Adverse Excursion allows us to use an alternative approach to evaluate a prospective Stop Loss level.

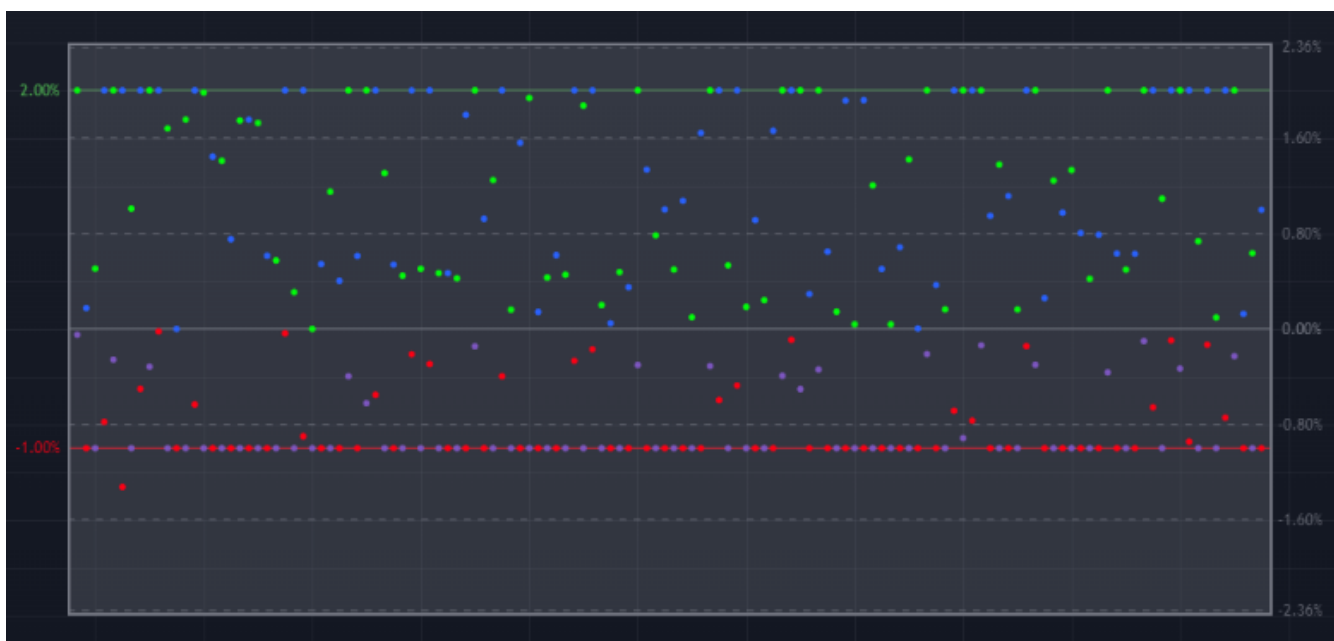
When developing a trading system, there are many factors that lead us to determine our Stop levels (if you like to have one), such as the asset's volatility, a percentage of our total capital, or a combination of various factors. If we express our risk as R and seek a risk-to-reward ratio of 2:1, we can say that in a winning trade, we will obtain $2R$. Let's take a look at an example using this concept:

If over a significant sample of trades the average profit is $1.5R$, but the MFE of this sample is 2.6, this may indicate that we are leaving profits on the table that we could capture by adjusting our Take Profit level.

Similarly, if we study the MAE of our system and consistently see that it is less than $1R$, we are in a case where we could potentially shorten our Stop. This can improve returns even using the same Take Profit.

But also, Maximum Adverse Excursion and Maximum Favorable Excursion can be used to determine a level from which we could close a trade and immediately open another in the opposite direction, have dynamic control over [position size](#) by increasing it while having full control of our risk, or modify our entry point into the market and improve our trades from scratch.

Here's an example of MAE and MFE in the analysis of a simple moving average crossover system:



Maximum Favorable Excursion

Desc.	LONG		SHORT		OVERALL	
	MFE	MAE	MFE	MAE	MFE	MAE
Min	0.00%	-1.33%	0.00%	-1.00%	0.00%	-1.33%
Mean	1.24%	-0.81%	1.13%	-0.81%	1.19%	-0.81%
Median	1.22%	-1.00%	1.22%	-1.00%	1.22%	-1.00%
Max	2.00%	-0.02%	2.00%	-0.05%	2.00%	-0.02%
Trades	66		66		132	
>= 2.00%	24		19		43	
<= -1.00%	43		47		90	

mfe mae

Maximum Adverse and Favorable Excursions are usually represented in a scatter plot where we can see, in this case as a percentage, their values compared to the Stop Loss and Take Profit.

The table shows useful data such as MAE and MFE by type of operation, whether Long or Short, Maximums, Minimums, Median, etc. Each trader determines which data is relevant when analyzing their system and how they express it.

To calculate the MAE, it's necessary to have a significant sample of data, since otherwise, the results may not be representative and may not allow for an accurate evaluation of the risk and profitability of a trade.

We made use of Maximum Adverse Excursion in an article where we showed a [leveraged trading strategy](#).

If the sample size is too small, the MAE may underestimate the actual risk of a trade and not provide a complete picture of the trading system's performance. Additionally, a small sample may not include extreme cases relevant to understanding a trading strategy's behavior in different market conditions.

Therefore, having a significant sample of data is essential to obtain an accurate evaluation of the MAE and make informed decisions about the profitability and risk of a trading strategy. It's important to note that the required sample size may vary depending on the trading strategy and the market conditions in which it operates, but in general, the larger the sample, the more reliable the MAE calculation will be.

How much MAE is good?

There isn't a specific MAE value that is generally considered "good" or "bad," as this depends on the specific context of each model and the data being analyzed.

In general, a model with a low MAE is considered to have higher accuracy than a model with a high MAE.

However, to determine whether the MAE of a model is good or not, it's necessary to compare it with other models trained with the same data and with the same prediction task.

Additionally, evaluating the performance of a model is not limited to a single MAE value, but other metrics should be considered, and appropriate statistical tests should be conducted to ensure that the model is reliable and generalizes well to unseen data.

The main problem for many traders, when they start live trading, is that they can't tolerate losses. Any strategy with a high MAE might lead to abandonment or selling at bottoms. In general, traders are intolerant to inevitable losses due to trading biases.

- [Systematic trading biases in trading](#)

Maximum Adverse Excursion (MAE) and Maximum Favorable Excursion (MFE) conclusions

The use of Maximum Adverse Excursion (MAE) and Maximum Favorable Excursion (MFE) can improve a trading system in several ways:

- 1. Evaluating the risk-reward relationship: MAE and MFE allow traders to evaluate the risk-reward relationship of a trade. If the MAE is significantly greater than the MFE, then the trader is risking much more compared to what they could potentially gain, suggesting that a review of the exit strategy is needed.
- 2. Establishing more effective loss limits: With the use of MAE, traders can adjust loss limits to reduce risk in a trade. If the MAE is too large compared to the established loss limit, then the trader may consider adjusting the loss limit to limit the risk.
- 3. Identify additional profit opportunities: With MFE, traders can identify additional profit opportunities during a trade. If the MFE is significantly greater than the profit realized in the trade, then the trader may consider adjusting the exit strategy better to take advantage of profit opportunities during the trade.
- 4. Improving Risk Management: The use of MAE and MFE can help [traders better manage their risk](#) and reduce the possibility of losing large amounts of money. By setting effective loss limits and taking advantage of additional profit opportunities, traders can improve their risk management and reduce their exposure to significant losses.

In summary, MAE and MFE can be useful tools for evaluating the performance of a trading system and making adjustments to improve its effectiveness. By using these indicators, traders can better manage their risk, identify profit opportunities, and establish effective loss limits to improve their overall trading performance.

FAQ:

Who introduced the concept of Maximum Adverse Excursion (MAE) in trading?

John Sweeney, the Technical Editor of Technical Analysis of Stocks and Commodities magazine, introduced the concept of Maximum Adverse Excursion. It is designed to assist traders in evaluating drawdown levels at which trades typically do not recover.

How much MAE is considered good in trading?

There isn't a specific "good" or "bad" value for MAE as it depends on the context of each trading model and the data being analyzed. Generally, a lower MAE is considered indicative of higher accuracy, but it should be compared with other models for a comprehensive assessment.

Why is having a significant sample size crucial when calculating MAE?

A significant sample size is essential to obtain an accurate evaluation of MAE. A small sample may not be representative of a trading strategy's performance and may not include extreme cases relevant to understanding its behavior in different market conditions.



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I've got an Msc from Heriot-Watt University, Edinburgh (1996), in addition a to a business administration degree the Norwegian School of Management (BI – 1994). Did my mandatory military service in between.

After university, I worked two years as an auditor (1996-1998).

I co-founded Aksjeforum.com in 1998/99 - one of the first websites about trading and investing in Norway. It was later acquired by Digi.no in 2001.

I have written 4 books about trading (in Norwegian). One of them has sold 30,000 copies, a record for a financial book in Norway.

From 2001 until 2018 full-time independent prop trader (Series 7 in 2001) and investor.

2018-today: Investor, writer, analyst.



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